

SESSION-05
SECURITIES ANALYSIS & PORTFOLIO MANAGEMENT

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

"By the end of these 6 sessions, you should be able to open any mutual fund fact sheet and know exactly which of these six questions its marketing is quietly avoiding. Which one would you guess — and why?"

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If Investors Hold Efficient Portfolios, How Should Securities Be Priced?

- Lineage: Markowitz (1952, efficient frontier) → Tobin (1958, adds R_f /CML) → Sharpe/Lintner (1964–65, CAPM equilibrium)
- This slide is the pivot: from portfolio-level pricing (CML) to security-level pricing (CAPM/SML)
- Every assumption listed here will be directly challenged later in this session



Full CAPM assumption set: *mean-variance optimizers · homogeneous expectations · unlimited borrowing/lending at R_f · no taxes/transaction costs · all assets liquid & divisible*

REFLECTION POINT

“Which of these five CAPM assumptions do you think is closest to reality in Indian markets today — and which is furthest from it?”

QUESTION BANK LINK

None — framing slide.

CAPM: The Model That Conquered Finance

- Directly connects to the Characteristic Line (Session 2, Slide 20): CAPM is the expected, equilibrium counterpart of that estimated, historical regression
- Gives a required-return benchmark against which actual return is compared — the alpha logic from Session 2, Slide 19

$$E(R_i) = R_f + \beta_i \times [E(R_m) - R_f]$$

R_f risk-free rate β_i systematic risk exposure market risk premium

HIGH-BETA

14.8%

$\beta = 1.4$

$R_f=5\%, R_m=12\%$

LOW-BETA

9.2%

$\beta = 0.6$

$R_f=5\%, R_m=12\%$

Market risk premium is not a fixed constant — it varies by period and geography (historically $\approx 5\text{--}7\%$ for equities globally) and is itself a contested figure.

REFLECTION POINT

“The market risk premium is often assumed to be a fixed 5–7%. What happens to every required-return calculation in this course if that number is actually wrong by even 1–2 percentage points?”

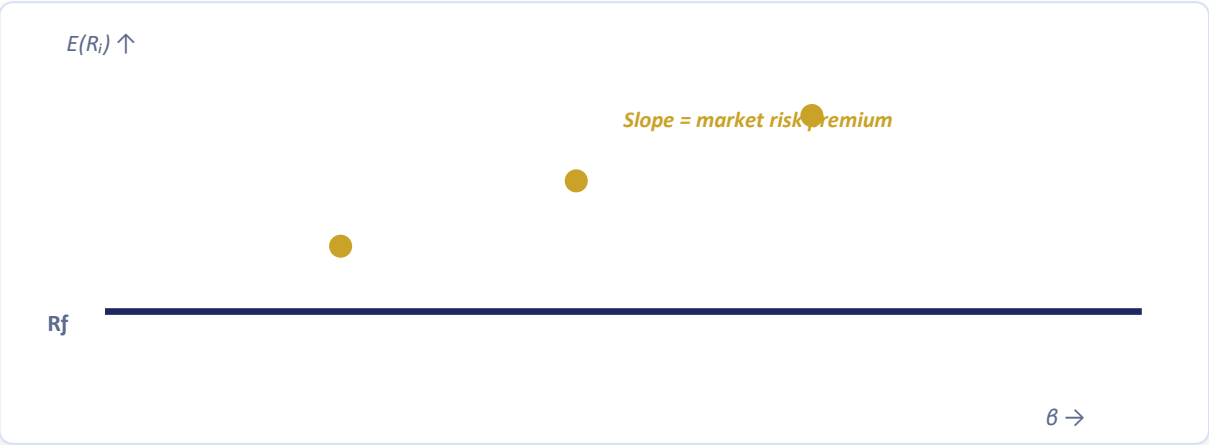
QUESTION BANK LINK

Use in class: Problem 7.1 (computing a point on the SML)

Assign for practice: Revisit Problem 4.6 (CAPM required return foundation, Session 2)

The Security Market Line: Graphing Required Return

- Plot $E(R_i)$ against β_i : a straight line through $(0, R_f)$ with slope equal to the market risk premium
- Every security's required return sits on this line — regardless of whether it is efficiently diversified



	CML	SML
Risk measure	Total risk (σ)	Systematic risk (β)
Applies to	Efficient portfolios only	Any security or portfolio

REFLECTION POINT

“A single stock cannot sit anywhere useful on the CML, but it can sit meaningfully on the SML. Why exactly does that difference exist?”

QUESTION BANK LINK

Problem 7.1 (reinforced from Slide 43) — no new problem introduced here.

Overvalued, Undervalued, and Fairly Valued Securities

- Above the SML: positive alpha, actual return exceeds required — undervalued / BUY signal
- Below the SML: negative alpha, actual falls short of required — overvalued
- A low-beta (defensive) security can be correctly priced even with lower expected return — “low return” ≠ “overvalued”



Stock A: 16% actual vs 14.8% required

$\alpha = +1.2\%$ — undervalued / BUY

Stock B: required 9.2%

Fairly priced for a defensive/liability-matching strategy despite lower return

REFLECTION POINT

“A client complains that a low-beta fund ‘never even hits the market’s return.’ How would you explain, using the SML, why that fund might still be perfectly and fairly priced?”

QUESTION BANK LINK

Use in class: Problem 7.2 (identifying an undervalued security)

Assign for practice: Problems 7.3 (overvalued), 7.4 (fairly valued defensive security)

Comparing Two Securities Using the SML

- The security with the higher expected return is not automatically the better investment
- The better investment is the one with the higher alpha after adjusting for beta risk — the classic exam/interview trap
- Should CAPM be supplemented with multi-factor screens (size/value, momentum)? CAPM as a floor, not a ceiling, of analysis

SECURITY X

$$E(R) = 13\%, \beta = 1.2$$

Required = 12.4%

$\alpha = +0.6\%$

SECURITY Y — RECOMMENDED

$$E(R) = 11\%, \beta = 0.7$$

Required = 9.5%

$\alpha = +1.5\%$ (higher alpha despite lower return)

REFLECTION POINT

“If Security Y has a lower expected return than Security X but a higher alpha, which one would you actually recommend to a client — and how would you justify it to someone who only looks at the headline return number?”

QUESTION BANK LINK

Use in class: Problem 7.5 (comparing two securities using the SML)

Roll's Critique: Is CAPM Even Testable?

- Roll (1977): the true market portfolio must include all risky assets globally — equities, bonds, real estate, private equity, human capital
- Any index (Nifty 50, S&P 500, MSCI World) is a flawed, incomplete proxy for this theoretical construct
- Switching a manager's benchmark from domestic to global can flip measured alpha from positive to negative for the same portfolio

NIFTY 50

Required = 10.4%

$\beta = 0.96$

Same stock, same day — different answer

MSCI WORLD

Required = 12.9%

$\beta = 1.30$

Same stock, same day — different answer

S&P 500

Required = 11.75%

$\beta = 1.15$

Same stock, same day — different answer

Same stock, $R_f=6.5\%$, market premium=6% — only the benchmark index changes.

REFLECTION POINT

“If beta and required return change this much depending on which benchmark you choose, has CAPM actually ever been properly tested — or is Roll right that it can't be?”

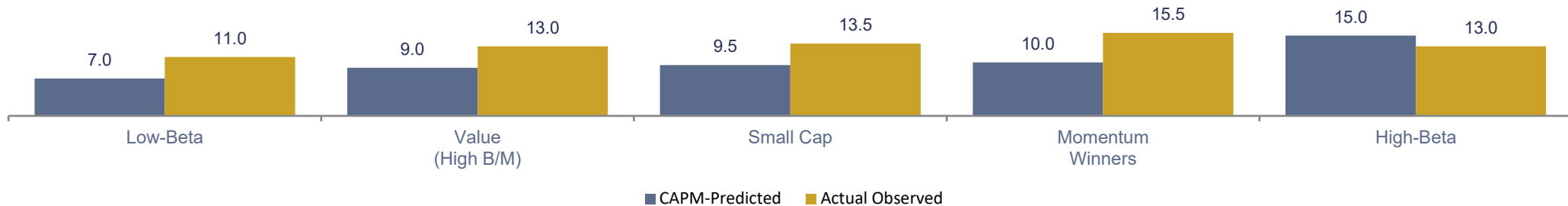
QUESTION BANK LINK

None — conceptual slide, directly reinforcing Session 2's Characteristic Line estimation-issues discussion.

Why CAPM Struggles in the Real World: Empirical Anomalies

- Low-beta anomaly: low-beta stocks persistently outperform CAPM's prediction
- Size effect (SMB) and Value effect (HML): small-cap and high book-to-market stocks earn persistent excess returns (Fama-French)
- Momentum effect: 12-month price winners continue to outperform (Carhart, 1997)

CAPM-PREDICTED VS ACTUAL OBSERVED RETURN (ILLUSTRATIVE, %)



REFLECTION POINT

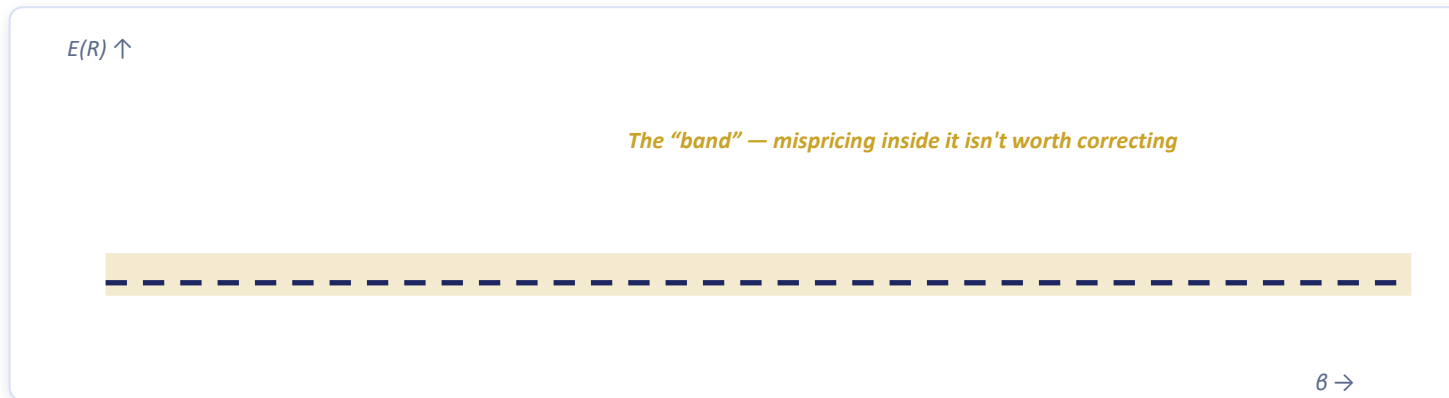
“Should CAPM be retired as the primary asset-pricing model given this evidence — or does it remain pedagogically and practically useful regardless of these anomalies?”

QUESTION BANK LINK

None — empirical-evidence slide, directly reinforcing Session 2's Slide 18 beta-debate discussion.

Transaction Costs, Behavioral Biases, and the SML “Band”

- Transaction costs transform the precise SML into a band, not a single line — mispricing smaller than the bid-ask spread plus execution costs isn't worth arbitraging
- Behavioural biases (loss aversion, overconfidence, herding) cause systematic deviations — especially pronounced in retail-dominated markets like India
- Real limit to diversification: transaction/monitoring costs eventually eclipse further marginal risk reduction — a limit pure theory ignores



WHY THE LINE BLURS

Costs + behaviour mean small anomalies can persist indefinitely without being irrational to leave uncorrected.

REFLECTION POINT

“If mispricing smaller than the transaction-cost band simply persists, is that a market failure — or is it actually rational behaviour by investors who correctly judge the trade not worth making?”

QUESTION BANK LINK

None — frictions and behavioural extension of the SML.

Session 5 Synthesis: Elegant Theory, Messy Practice

- The debate is deliberately not resolved here — it returns with a full structured classroom debate in Session 6
- CAPM remains the dominant model taught in corporate finance and used by most CFOs for capital budgeting, despite everything raised this session

CAPM'S LOGIC

$E(R_i) = R_f + \beta_i \times \text{premium}$ — clean equilibrium logic, direct link to the Characteristic Line, a real buy/sell/hold valuation payoff.



CAPM'S CHALLENGES

Roll's Critique, empirical anomalies (low-beta, size, value, momentum), transaction-cost frictions, behavioural biases.

Deliberately unresolved — Session 6 → Risk-Adjusted Performance Evaluation

REFLECTION POINT

"If CAPM is this contested, why does it remain the dominant model taught in corporate finance and used by most CFOs for capital budgeting?"

QUESTION BANK LINK

None — pure bridge/synthesis slide closing Session 5.